AI.406

Robot Perception

COURSE OVERVIEW

Advanced robot perception: multimodal fusion, deep perception, and perception for autonomous operation.



ILLUSTRATIVE HERO IMAGERY

ROBOT PERCEPTION

1

SESSION

60 MINUTES

ORIENTATION: SEEING TO ACT

KEY TOPICS

- Robot perception stack
- Multimodal perception
- Perception under uncertainty
- Applications

TEACHING FOCUS

Fuse senses into understanding.

KEY TAKEAWAY Robots perceive with many senses at once.

ROBOT PERCEPTION

2

SESSION

60 MINUTES

DEEP PERCEPTION

KEY TOPICS

- CNNs for robot vision
- 3D deep learning
- Perception networks
- Real-time deep perception

TEACHING FOCUS

Use deep networks to perceive.

KEY TAKEAWAY Deep perception sees richly and quickly.

3

SESSION

60 MINUTES

TACTILE & FORCE SENSING

KEY TOPICS

- Tactile sensors
- Force-torque sensing
- Slip and texture
- Tactile servoing

TEACHING FOCUS

Feel what the robot touches.

KEY TAKEAWAY Touch completes vision.

ROBOT PERCEPTION

4

SESSION

60 MINUTES

PERCEPTION FOR AUTONOMY

KEY TOPICS

- Obstacle and free space
- Semantic perception
- Human detection
- Perception–action loop

TEACHING FOCUS

Perceive to navigate and act.

▲ **KEY TAKEAWAY** Autonomous perception drives action.

ROBOT PERCEPTION

5

SESSION

60 MINUTES

PERCEPTION SYSTEMS

KEY TOPICS

- Sensor calibration
- Data association
- Real-time pipelines
- Perception validation

TEACHING FOCUS

Engineer robust perception systems.

KEY TAKEAWAY Perception must work in the wild.

ROBOT PERCEPTION

6

SESSION

60 MINUTES

ROBOT PERCEPTION & REVIEW

KEY TOPICS

- Perception demo
- Course recap and assessment
- Toward robot navigation (AI.407)
- Perception eval

TEACHING FOCUS

Integrate perception on a robot.

KEY TAKEAWAY Perception closes the loop to action.